

ASTHA ERP

Enterprise-Grade Cloud-Enabled ERP for Astha Builders & Hardware

System Specification & Blueprint Document

Target Platforms: Windows Desktop (Offline-First) + Web Admin + Mobile Companion

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Document Overview & Table of Contents

This document serves as the primary system specification, database catalog, and design blueprint for ASTHA ERP, a custom enterprise solution designed specifically for Astha Builders & Hardware. It details all requirements for building an offline-first system capable of scaling to future Android/iOS releases, with seamless cloud synchronization and robust double-entry accounting integrity.

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1. Executive Summary & Vision

Astha Builders & Hardware operates a high-volume retail and wholesale distribution network for construction materials, building supplies, hardware items, and paint products. With multiple warehouses, delivery yards, and high-frequency counter sales, the business faces complex logistical and financial operations.

Core Challenges & Business Needs

- **Connectivity Constraints:** Retail yards and warehouses are frequently located in areas with intermittent internet access. If internet connectivity drops, counter billing and stock movements must continue without interruption.
- **Inventory Complexity:** Items vary widely (bulk cement bags, steel bars in bundles, liquid paint in batches with expiry, loose nuts and bolts). Warehouse transfers and automated reorder alerts are essential to avoid project supply disruptions.
- **Accounting Integrity:** Operations span cash sales, high-value credit sales, partial customer collections, and bulk supplier credits. Ad-hoc accounting sheets fail to prevent revenue leakage. The system requires a strict double-entry ledger engine.
- **Regulatory Compliance:** Tax compliance requires error-free GST invoicing, calculation of CGST, SGST, IGST, and generation of GSTR reports (GSTR-1, GSTR-3B) with active HSN/SAC code cataloging.

The Unified Solution

ASTHA ERP is designed as a hybrid platform that solves these problems through a distributed system:

- **Desktop App (EXE):** Installed at retail counters. It operates on a local database (SQLite) for extreme speed and offline independence. All sales invoices, cash collection receipts, and inventory picks are saved locally.
- **Cloud Server (FastAPI + PostgreSQL):** Hosts the master data. Serves as the central clearinghouse for inventory reconciliation, consolidated accounting, and owner-facing reports.
- **Sync Engine:** Runs quietly in the background on the desktop clients. When connectivity is available, it pushes local transaction changesets and pulls catalog updates, handling data conflicts deterministically.

2. Technology Stack & Deployment

To ensure a cross-platform, long-term supportable system, the software stack uses modern, performant, and memory-safe tools across three primary vectors.

Layer	Technology Selected	Purpose & Rationale
Desktop Frontend	React 19 + TypeScript + Vite + TailwindCSS	Responsive, interactive, easy component reuse across desktop and web applications.
Desktop Wrapper	Tauri v2 + Rust Process Management	Generates lightweight (under 20MB), high-performance Windows executables (EXE) with direct printer access.
Local Database	SQLite 3	Extremely fast embedded database requiring zero configuration, storing offline transactions.
Web Portal	Next.js 15 (React 19) + TailwindCSS	Provides a cloud-hosted, zero-install admin dashboard for multi-branch monitoring and accounting.
Backend API	FastAPI (Python 3.12 / 3.14)	Asynchronous, high-performance web framework for syncing, reporting, and integration scripts.
Cloud Database	PostgreSQL 16	Enterprise relational database with robust row-level security and JSONB indexing for audit trails.
Sync Transport	WebSockets + HTTP Delta Payloads	Ensures instant messaging for active connections, falling back to HTTP batches during sync recovery.
ORM & Migrations	SQLAlchemy + Alembic	Ensures uniform database queries and simple schema versioning across dev and production envs.

Offline-First Sync Engine Specification

The system enforces a write-locally, sync-globally data flow. The local desktop app reads and writes exclusively to the local SQLite database. A dedicated sync worker operates in a separate Rust thread within Tauri:

1. **Change Queue:** Every write operation (e.g., creating a sales invoice) inserts a row into a `sync\_queue` table containing the table name, action type (INSERT, UPDATE), timestamp, and a JSON payload of the changes.

2. **Push Sequence:** When internet is active, the sync client pushes local queue entries in order of transaction timestamp. The cloud server validates the transaction, applies business rules, and saves it to PostgreSQL.

3. **Pull Sequence:** The cloud server broadcasts catalog updates (new items, changed prices, supplier credit revisions) using WebSockets or SSE (Server-Sent Events) to listening desktop nodes.

4. **Conflict Resolution Protocol:**

- *Ledger/Invoices:* Invoices are assigned unique alphanumeric serial codes prefixed by the local branch identifier. Since invoice sequences are distinct per branch, duplicate primary keys are avoided. If edits collide, Last-Write-Wins (LWW) is enforced by comparing record-level update timestamps.

- *Stock Levels:* Real-time inventory is tracked using raw stock movement cards (debit/credit records). Instead of overwriting total stock quantities, the sync engine applies the delta shift (e.g., subtracting 10 bags of cement). This allows concurrent offline sales at different branches to merge without corrupting totals.

3. Project Directory Architecture

The ASTHA ERP codebase is organized as a monorepo, separating application runtimes while sharing core logic, validation schemas, and database types. This layout ensures maximum code reusability and minimizes drift between the Web and Desktop builds.

Directory Path	Domain	Contents & Purpose
/apps/desktop	Desktop client	Tauri core configurations, window handlers, serial/USB print modules, SQLite hooks.
/apps/web	Cloud Web portal	Next.js admin portal, reporting dashboard, configurations panel, multi-branch view.
/packages/shared	Shared code	TypeScript definitions, common interfaces, mathematical discount/tax validators, regex utilities.
/packages/db-schema	Database logic	Alembic migrations, standard tables definitions, local SQLite structure definitions.
/services/api	Cloud Backend	FastAPI app code, database routers, tax engines, report compilers, SMS/WhatsApp engines.
/services/sync-server	Sync Engine	WebSocket hubs, conflict resolution logic, queue consumers, network status health monitors.
/scripts	Maintenance	Automated backup cron scripts, security audits, database seeders for testing.

Detailed Microservice Architecture Layout

The diagram below illustrates the communications between the monorepo layers and dependencies:

Local Counter POS (Tauri EXE)	Sync Transport	Cloud Infrastructure
Tauri Webview UI (React) ↓ (Local IPC) Tauri Rust Backend ↓ (SQL queries) SQLite Local File	HTTP POST /sync WebSockets (Real-time events) AES-256 Encrypted payloads Delta sync changesets	FastAPI Gateway ↓ (Async requests) Cloud Workers / Task Queue ↓ (SQLAlchemy) PostgreSQL Database ↓ (Cloud Storage) AWS S3 (Daily Backups)

4. Unified Database Catalog

The tables below detail the foundational PostgreSQL (Cloud) and SQLite (Local) database structures. All primary keys are UUIDs generated by the client to support offline creation without server-side sequence allocation. Monotonically increasing numbers (like invoice IDs) use client-side unique prefixes.

4.1 Authentication & User Access Tables

Table Name: `roles` (System Role Definitions)

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY	Unique ID for the role (1 = Admin, 2 = Manager, 3 = Accountant, 4 = Sales, 5 = Store, 6 = Cashier).
name	VARCHAR(30)	NOT NULL, UNIQUE	The name of the role (e.g., 'Accountant').
description	TEXT	NULLABLE	Details of user responsibilities for this role.

Table Name: `users` (User Credentials & Status)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Unique client-generated ID.
username	VARCHAR(50)	NOT NULL, UNIQUE	Alphanumeric login identifier.
email	VARCHAR(100)	NULLABLE	Contact email address for recovery.
password_hash	VARCHAR(255)	NOT NULL	Password hash stored using Argon2id algorithm.
role_id	INT	FOREIGN KEY (roles.id)	Assigned security role mapping.
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE	Active flag. Set to false to disable access.
created_at	TIMESTAMP	NOT NULL, DEFAULT NOW()	Date and time the account was created.

4.2 Customer & Supplier Ledger Tables

Table Name: `customers` (Customer Directory & Constraints)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Customer UUID.
name	VARCHAR(100)	NOT NULL	Full name of the contact person.
company_name	VARCHAR(100)	NULLABLE	Name of the firm (e.g., 'Astha Builders').
phone	VARCHAR(15)	NOT NULL	Primary mobile number for WhatsApp alerts.
email	VARCHAR(100)	NULLABLE	Primary email address.
gstn	VARCHAR(15)	NULLABLE	15-character Indian Goods and Services Tax Identification Number.
credit_limit	NUMERIC(12,2)	DEFAULT 0.00	Maximum outstanding ledger balance allowed.
opening_balance	NUMERIC(12,2)	DEFAULT 0.00	Initial balance before using this system.

opening_type	VARCHAR(2)	DEFAULT 'DR'	DR (Debit / Receivable) or CR (Credit / Payable).
created_at	TIMESTAMP	DEFAULT NOW()	System insertion time.

Table Name: `customer\_ledgers` (Running Customer Ledger Cards)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Ledger entry UUID.
customer_id	UUID	FK (customers.id)	Link to target customer.
txn_date	DATE	NOT NULL	Date of transaction activity.
voucher_type	VARCHAR(10)	NOT NULL	Voucher class: 'SALE', 'RECEIPT', 'RETURN', 'OPENING'.
voucher_id	UUID	NOT NULL	UUID of matching document (Sales Invoice, Receipt Voucher, etc.).
debit	NUMERIC(12,2)	DEFAULT 0.00	Amount receivable from customer (increases outstanding balance).
credit	NUMERIC(12,2)	DEFAULT 0.00	Amount received from customer (reduces outstanding balance).
running_balance	NUMERIC(12,2)	NOT NULL	Calculated balance after transaction is posted.
remarks	TEXT	NULLABLE	Transaction description or invoice details.

Table Name: `suppliers` (Supplier Directory)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Supplier UUID.
name	VARCHAR(100)	NOT NULL	Vendor name or point of contact.
company_name	VARCHAR(100)	NOT NULL	Registered supplier business name.
phone	VARCHAR(15)	NOT NULL	Primary mobile number.
gstln	VARCHAR(15)	NULLABLE	Indian GST identification number.
opening_balance	NUMERIC(12,2)	DEFAULT 0.00	Initial balance on vendor ledger.
opening_type	VARCHAR(2)	DEFAULT 'CR'	DR (Receivable/Refund) or CR (Payable/Liability).

4.3 Inventory & Warehouse Stock Tables

Table Name: `items` (Product Catalog Catalog)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Product identifier.
name	VARCHAR(150)	NOT NULL	Name of product (e.g., 'Ultratech Cement OPC 53').
sku	VARCHAR(50)	UNIQUE	Stock Keeping Unit code.
barcode	VARCHAR(50)	NULLABLE	UPC, EAN, or custom system barcode for scanning.
category_id	INT	FOREIGN KEY	Product Category classification.
brand_id	INT	FOREIGN KEY	Product brand (e.g., 'TATA', 'Birla').
primary_unit_id	INT	FOREIGN KEY	Base stock unit (e.g., 'Bag', 'Box').
secondary_unit_id	INT	NULLABLE, FK	Sub-stock unit for retail splits (e.g., 'kg', 'Piece').
hsn_code	VARCHAR(8)	NOT NULL	Harmonized System of Nomenclature code for GST mapping.

gst_rate	NUMERIC(5,2)	NOT NULL	Integrated GST tax percentage (e.g., 18.00, 28.00).
reorder_level	NUMERIC(10,2)	DEFAULT 10.00	Minimum stock level threshold for alerts.
is_active	BOOLEAN	DEFAULT TRUE	Item availability toggle.

Table Name: `stocks` (Warehouse-Batch Mapping / Inventory State)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Unique stock batch identifier.
item_id	UUID	FK (items.id)	Link to catalog item.
warehouse_id	INT	FOREIGN KEY	Link to specific warehouse or yard.
batch_no	VARCHAR(50)	NOT NULL	Manufacturer batch number for tracking and audits.
expiry_date	DATE	NULLABLE	Product expiration date (critical for chemical paints/cements).
qty	NUMERIC(12,2)	DEFAULT 0.00	Physical stock count currently on hand.
purchase_rate	NUMERIC(12,2)	NOT NULL	Purchase cost per unit before GST.
sale_rate	NUMERIC(12,2)	NOT NULL	Target selling rate before GST.

4.4 Sales Invoicing & GST Logging Tables

Table Name: `sales\_invoices` (Sales Invoices Ledger)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Invoice UUID.
invoice_no	VARCHAR(30)	NOT NULL, UNIQUE	Formatted invoice number (e.g., 'AS-26-0001').
invoice_date	DATE	NOT NULL	Date of sale transaction.
customer_id	UUID	FK (customers.id)	Link to target customer.
warehouse_id	INT	FOREIGN KEY	Warehouse inventory source.
sub_total	NUMERIC(12,2)	NOT NULL	Sum of items net rates before taxes and discounts.
discount_amount	NUMERIC(12,2)	DEFAULT 0.00	Total discount deducted from gross value.
cgst	NUMERIC(12,2)	DEFAULT 0.00	Total Central GST collected (intrastate).
sgst	NUMERIC(12,2)	DEFAULT 0.00	Total State GST collected (intrastate).
igst	NUMERIC(12,2)	DEFAULT 0.00	Total Integrated GST collected (interstate).
total_amount	NUMERIC(12,2)	NOT NULL	Final invoice value (Subtotal - Discount + GST).
payment_mode	VARCHAR(20)	NOT NULL	Payment method: 'CASH', 'CREDIT', 'BANK', 'UPI', 'MIXED'.
payment_status	VARCHAR(15)	NOT NULL	Payment status: 'PAID', 'UNPAID', 'PARTIAL'.
sync_status	BOOLEAN	DEFAULT FALSE	Offline sync confirmation flag.

Table Name: `sales\_invoice\_items` (Invoice Line Items)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Line item ID.
invoice_id	UUID	FK (invoices.id)	Link to parent invoice.
item_id	UUID	FK (items.id)	Link to product catalog.
batch_no	VARCHAR(50)	NOT NULL	Selected item batch code.

qty	NUMERIC(12,2)	NOT NULL	Quantity purchased.
rate	NUMERIC(12,2)	NOT NULL	Selling rate per unit (excl. GST).
discount_percent	NUMERIC(5,2)	DEFAULT 0.00	Line discount rate percentage.
cgst_rate	NUMERIC(5,2)	NOT NULL	Central GST rate applied (e.g., 9.00).
sgst_rate	NUMERIC(5,2)	NOT NULL	State GST rate applied (e.g., 9.00).
igst_rate	NUMERIC(5,2)	NOT NULL	Integrated GST rate applied (e.g., 18.00).
total_amount	NUMERIC(12,2)	NOT NULL	Final line total (Qty * Net Rate + Tax).

4.5 Accounting System & General Ledger Tables

Table Name: `accounts` (Chart of Accounts)

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY	Ledger Account account code (e.g., 1001, 2005).
code	VARCHAR(20)	NOT NULL, UNIQUE	Chart of accounts grouping identifier.
name	VARCHAR(100)	NOT NULL	Account name (e.g., 'State Bank of India A/c').
parent_id	INT	NULLABLE, FK	Parent account ID for nested groups.
account_type	VARCHAR(20)	NOT NULL	Class: 'Asset', 'Liability', 'Equity', 'Revenue', 'Expense'.
opening_balance	NUMERIC(12,2)	DEFAULT 0.00	Day-zero ledger balance.
opening_type	VARCHAR(2)	DEFAULT 'DR'	DR (Debit) or CR (Credit) indicator.

Table Name: `vouchers` (General Vouchers Ledger)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Voucher ID.
voucher_no	VARCHAR(30)	NOT NULL, UNIQUE	Unique transaction reference code (e.g., 'JV-26-0001').
voucher_date	DATE	NOT NULL	Entry creation date.
voucher_type	VARCHAR(15)	NOT NULL	Class: 'JOURNAL', 'RECEIPT', 'PAYMENT', 'CONTRA'.
narration	TEXT	NULLABLE	Overall transaction description.
total_amount	NUMERIC(12,2)	NOT NULL	Voucher transaction total.

Table Name: `voucher\_items` (Accounting Double-Entry Split Items)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Double entry split item ID.
voucher_id	UUID	FK (vouchers.id)	Link to parent voucher.
account_id	INT	FK (accounts.id)	Target Chart of Accounts ledger.
debit	NUMERIC(12,2)	DEFAULT 0.00	Amount debited (must be 0.00 if credit has value).
credit	NUMERIC(12,2)	DEFAULT 0.00	Amount credited (must be 0.00 if debit has value).
narration	TEXT	NULLABLE	Row-level memo or notes.

4.6 Security Audit Logs Table

Table Name: `audit\_logs` (Internal Security & Transaction Trails)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY	Log record identifier.
user_id	UUID	FOREIGN KEY	User who performed the action.

action	VARCHAR(50)	NOT NULL	Action description: 'DELETE_INVOICE', 'EDIT_LEDGER', etc.
table_name	VARCHAR(50)	NOT NULL	Target database table affected.
record_id	UUID	NOT NULL	Target record identifier.
old_value	JSONB	NULLABLE	JSON snapshot of data state before change.
new_value	JSONB	NULLABLE	JSON snapshot of data state after change.
ip_address	VARCHAR(45)	NOT NULL	Origin IP address of user.
created_at	TIMESTAMP	DEFAULT NOW()	Action execution time.

5. Authentication & User Roles

ASTHA ERP implements strict role-based access control (RBAC). In offline desktop mode, session tokens are securely signed using a local private key and verified by the desktop app database. In online mode, operations query the FastAPI OAuth2 service to retrieve JWT (JSON Web Tokens).

System Role Definitions

- **Admin:** Has access to all system modules, global configurations, database back-ups, user creation, log reviews, and final accounting reconciliations.
- **Manager:** Supervises daily inventory levels, approves stock adjustments, oversees purchase order processes, and views standard client reports.
- **Accountant:** Enters journal vouchers, reviews tax filings (CGST, SGST, IGST), reconciles ledgers, and compiles financial statements (Trial Balance, P&L, Balance Sheet).
- **Sales Staff:** Creates sales quotations, converts quotations to sales orders, generates GST invoices, and tracks customer credit statuses.
- **Store Staff:** Updates stock receipts, processes warehouse transfers, prints barcodes, and logs item updates.
- **Cashier:** Manages daily point-of-sale billing, cash registers, counter returns, and registers receipts.

Granular Permissions Mapping Matrix

The table below details the accessibility matrix across primary system components:

Permissions Scope	Admin	Manager	Accountant	Sales	Store Staff	Cashier
System Setup / Config	Full Access	View Only	Denied	Denied	Denied	Denied
User Mgmt & Audits	Full Access	View Only	Denied	Denied	Denied	Denied
Inventory Levels Edit	Full Access	Full Access	View Only	Denied	Full Access	Denied
Sales Invoices View	Full Access	Full Access	Full Access	Full Access	View Only	Full Access
Sales Invoices Edit/Del	Full Access	Approve Only	Denied	Denied	Denied	Denied
General Ledger Postings	Full Access	Denied	Full Access	Denied	Denied	Denied
Reports View	Full Access	Full Access	Full Access	Own Only	Denied	Own Only
Backup & Restores	Full Access	Denied	Denied	Denied	Denied	Denied

Custom Permissions Setup

Managers can configure custom overrides. For example, a trusted Sales Associate can be granted temporary permission to override credit limits by inputting a manager's secure OTP or pin code at checkout.

6. Customer & Supplier Management

Managing trade terms, outstanding balances, and risk analysis metrics is central to hardware operations.

Customer Aging Analysis

To maintain liquidity, ASTHA ERP dynamically groups unpaid receivables into four aging columns. Aging is calculated relative to the current system date or selected cutoff date:

- **0 to 30 Days:** Current bills. Standard trade cycle.
- **31 to 60 Days:** Gentle alerts sent via SMS/WhatsApp.
- **61 to 90 Days:** Overdue. Requires manual collections follow-up.
- **90+ Days:** Delinquent. System alerts manager and blocks new credit sales.

Customer Name	Credit Limit	0-30 Days	31-60 Days	61-90 Days	90+ Days	Total Due
Astha Builders Ltd	500,000.00	120,000.00	80,000.00	0.00	0.00	200,000.00
Sri Balaji Cements	200,000.00	45,000.00	30,000.00	25,000.00	0.00	100,000.00
National Projects	300,000.00	0.00	0.00	50,000.00	85,000.00	135,000.00

Credit Limit Enforcement System

Whenever a sales order or invoice is compiled:

1. The client queries the customer's ledger to retrieve the current balance:

Balance = Sum(Debits) - Sum(Credits)

2. If the balance exceeds the customer's `credit\_limit`, a warning banner is displayed.
3. If `is\_strict\_limit` is enabled, the system prevents checkout. The salesperson must select an alternative payment method (Cash, Bank) or request an administrative OTP override.

Supplier Payments & Balances

Suppliers use credit accounts. Purchase receipts generate credits (liabilities) on the supplier's ledger, and cash or bank payment vouchers post debits (reducing the liability). Payment schedules are color-coded in the dashboard to ensure payments are made on time.

7. Inventory & Warehouse Management

Building materials require detailed stock keeping units (SKUs) and batch management to track physical locations and prevent item spoilage.

Batch / Lot & Expiry Specifications

Certain materials, such as cement and chemical paints, have shelf lives. The system handles batches as follows:

- **FIFO Stock Picking:** POS checkout screens recommend items from the oldest unexpired batch to ensure inventory moves efficiently.
- **Expiry Notifications:** The system monitors batch expiration dates (`expiry\_date`) and alerts managers 30 days before expiration, flagging items to be discounted or returned to vendors.

Multi-Unit Conversions Engine

Items are often purchased in bulk and sold in retail splits. This is managed through unit conversion factors:

- **Example:** Cement is purchased in Boxes (containing 12 bags each) and sold in individual Bags.
- **Formula:** *Quantity in Sub-Unit = Bulk Quantity * Conversion Factor*

Item Name	Purchase Unit	Sales Unit	Conversion Factor	System Quantity on Hand
Nippon Paint Emulsion	Box (4 Tins)	Tin	4.00	12 Boxes (48 Tins)
TMT Steel Rods 12mm	Bundle (10 Pcs)	Piece	10.00	8 Bundles (80 Pieces)
General Purpose Nails	Box (25 kg)	kg	25.00	2 Boxes (50 kg)

Warehouse Transfer Protocol (WTP)

When moving goods between the main warehouse and retail yards:

1. A **Warehouse Transfer Voucher (WTV)** is created in a 'Draft' state by the dispatching warehouse, deducting items from its local virtual stock count.
2. A **Transit Registry** tracks the items in transit.
3. The receiving warehouse inspects the delivery and clicks 'Confirm Receipt' on their terminal, adding the verified items to their local stock count and closing the transit record. Discrepancies are logged in an audit report.

8. Sales, Billing & GST Invoice Engine

The sales module handles billing, quotations, and returns. POS systems require keyboard-only workflows to keep counter lines moving quickly.

Standard POS Keyboard Bindings

The desktop application supports keyboard shortcuts to speed up billing:

- **[F1]:** Focus search bar to scan or type item names.
- **[F2]:** Edit quantity of the selected line item.
- **[F3]:** Apply discount percentage or cash amount to invoice.
- **[F5]:** Save invoice to local database, queue for sync, and print.
- **[F8]:** Open quick receipt panel for customer credit cash collections.
- **[ESC]:** Clear current draft invoice.

GST Calculation Specifications

Tax is computed at the line-item level using HSN codes to ensure accuracy during partial returns:

- **Intrastate Sales (Billing Address state == Shipping Address state):** CGST and SGST are calculated as 50% of the item's GST rate each.
- **Interstate Sales (Billing Address state != Shipping Address state):** IGST is calculated at 100% of the item's GST rate.

Formulas:

*Taxable Amount = (Quantity * Rate) - Line Discount*

*CGST = Taxable Amount * (GST % / 200)*

*SGST = Taxable Amount * (GST % / 200)*

*IGST = Taxable Amount * (GST % / 100)*

Line Total = Taxable Amount + CGST + SGST + IGST

Invoice Printing Modes

Desktop nodes print to two targets:

1. **A4/A5 Laser Prints:** Generates professional GST invoices for contractors, detailing tax brackets, bank accounts, and terms.
2. **3-Inch Thermal POS Receipt:** Quick receipt format for cash sales, listing items, totals, and a payment QR code.

9. Double-Entry Accounting Core

A strict accounting engine underpins ASTHA ERP. Unlike simple ledgers, every inventory movement or payment must generate balanced journal entries across the Chart of Accounts.

Chart of Accounts Organization

Accounts are organized in a nested tree structure. The root groups are: Assets, Liabilities, Equity, Revenues, and Expenses.

Account Code	Account Name	Root Group	Standard Balance Type
1000	ASSETS	Asset Root	Debit
1001	State Bank of India A/c	Current Asset	Debit
1002	Cash In Hand	Current Asset	Debit
1003	Trade Receivables (Customers)	Current Asset	Debit
2000	LIABILITIES	Liability Root	Credit
2001	Trade Payables (Suppliers)	Current Liability	Credit
2002	Duties & Taxes (GST Output)	Current Liability	Credit
3000	REVENUES	Revenue Root	Credit
3001	Hardware Sales Income	Direct Income	Credit
4000	EXPENSES	Expense Root	Debit
4001	Purchase Account	Direct Expense	Debit
4002	Freight & Transport Outward	Direct Expense	Debit

Voucher Processing Rules

The system enforces the following transaction rules:

- Debits equal Credits:** No voucher can be saved unless the sum of its debits matches the sum of its credits.
- Contra Vouchers:** Contra entries are restricted to cash-to-bank deposits or bank-to-cash withdrawals (e.g., Cash In Hand A/c and SBI Bank A/c). Other ledgers are disabled in this view.
- Journal Vouchers:** General adjustments, depreciation, and bad debts are logged via Journal Vouchers. Cash/Bank accounts are typically excluded from general journals to prevent cash leakage.
- Automatic Integration:** Creating a Sales Invoice automatically triggers background postings to: Debit Customer A/c, Credit Sales Revenue A/c, and Credit CGST/SGST Liability Accounts.

10. Integrations & Utility Engines

External integrations automate notifications, labels, and backups, extending the system beyond a simple database application.

WhatsApp & Email Notification Service

When a customer invoice or payment receipt is saved:

- 1. The backend triggers a worker process that generates a secure, read-only PDF version of the invoice.
- 2. The worker sends the document to the customer's phone number using the WhatsApp Business API.
- 3. An email is sent simultaneously to the client's inbox with the PDF statement attached.

Automated Backup Engine

To prevent data loss, the backup engine uses a dual-destination strategy:

- **Local Node Backups:** In desktop mode, a scheduled task creates an encrypted database backup on a local directory (or external USB drive) every hour.
- **Cloud Backups:** The cloud database runs daily automated logical backups (``pg_dump``). Backups are encrypted using AES-256 and uploaded to a secure AWS S3 bucket, with a 30-day retention policy.

Barcode & QR Label Printing Specifications

Barcode labels for un-scannable inventory items (e.g., loose pipes, custom bricks) are printed using TSC or Zebra printers via direct raw print commands. The system generates Code 128 barcodes along with price details.

Line	Content / Command	Variable Mapped	Sample Output
1	BARCODE 20,20,"128",40,1,0,2,2	item.barcode_value	1002049283
2	TEXT 20,70,"ROMAN.F",0,1,1	item.display_name	OPC 53 Cement - Astha
3	TEXT 20,90,"ROMAN.F",0,1,1	item.sale_price + taxes	MRP: Rs. 450.00 (Incl. Tax)

11. QA & Implementation Plan

To build a stable platform, development is split into milestones, starting with core databases and ending with multi-device synchronization testing.

QA Testing Strategy

- **Unit Tests:** Focus on accounting equations, GST calculation methods, unit conversion formulas, and sync queue serialization.
- **Integration Tests:** Verify local SQLite write operations and sync queue transfers to the FastAPI server, simulating network drops to ensure data consistency.
- **E2E Tests:** Use Playwright to simulate sales checkouts, cashier shifts, and customer payment processing.

Development Milestones & Roadmap

The project is planned over an intensive 16-week cycle:

Phase	Focus Area	Key Deliverables	Target Timeline
Milestone 1	Core Infrastructure	PostgreSQL schema, SQLite migrations, Tauri packaging, User Auth APIs.	Weeks 1-2
Milestone 2	Sync Engine	Delta tracking queue, WebSocket state hub, conflict merge models.	Weeks 3-4
Milestone 3	Inventory Module	Brands, Categories, Batch/Expiry controls, stock transfers.	Weeks 5-6
Milestone 4	Sales & GST	Point of sale (POS) billing interface, thermal prints, tax engines.	Weeks 7-8
Milestone 5	Accounting	Chart of accounts, Journal/Contra vouchers, Double entry logic.	Weeks 9-10
Milestone 6	Reporting Suite	Trial Balance, Balance Sheet, GST reports, Excel exports.	Weeks 11-12
Milestone 7	Integrations	WhatsApp API, ESC/POS barcode prints, S3 backups.	Weeks 13-14
Milestone 8	QA & Deployment	Production builds (Windows EXE, Docker web files), security audits.	Weeks 15-16

System Initialization Guide

To run the PDF compilation script or start database seeding:

1. Run ``pip install reportlab`` to install the PDF generator library.
2. Run ``python generate_spec.py`` to compile the PDF specification document.
3. The compiled document will be saved in the project root directory as: ``ASTHA_ERP_Master_Specification.pdf``.